

Cluster origins — the cross-domain appendix

- **Run:** 12 September 2026, one sitting.
- **Method:** the `lit-review` skill ([reproducible-lit-review](#)), stages 1–6, under the stopping rule below.
- **Related:** [experiment register.md](#) · [peer review mapping.md](#) · [literature review plan.md](#)
- **Status:** complete for this sitting. One register amendment follows from it (§ "The register amendment"); four mechanism candidates are proposed, none adopted.

Stopping rule, written before reading anything. 15–20 sources, one sitting. Review articles, handbook chapters and textbook overviews only. Nothing here enters the 58-paper shortlist, gets Elicit extraction, or reopens the funnel. Output is at most one paragraph per cluster plus a list of concrete implications.

Held to. 16 sources shortlisted from a pool of 611, every one of type `review`, `book-chapter` or `book`. Extraction is **abstract-only** — 1 of 16 sources yielded a PDF, because books and chapters have no open full text.

Pool (backward + forward + name search)	611
Excluded on source type	362 <code>article</code> , plus 20 others
Excluded below 2 cluster keywords	193
Passed screening	31
Shortlisted	16
PDFs obtained	1 of 16 (6.3%)
Verdicts	10 <code>nothing</code> · 3 <code>supports</code> · 3 <code>challenges</code>

Ten of sixteen sources contributing nothing is the outcome the stopping rule named as good: *"A pass where three of four clusters say 'background only' is a perfectly good outcome, and much better than a set of post-hoc justifications."*

1. Seeds, and what the corpus turned out to be

Seed	Year	Cluster	Backward refs
Gibson, <i>The Ecological Approach to Visual Perception</i>	1979	D	0
Chemero, <i>An Outline of a Theory of Affordances</i>	2003	D	35
Gopnik et al., <i>A Theory of Causal Learning in Children</i>	2004	A	151
Spelke & Kinzler, <i>Core knowledge</i>	2006	A	92
Conant & Ashby, <i>Every good regulator...</i>	1970	C	3
Powers, <i>Feedback: Beyond Behaviorism</i>	1973	E	6

All six resolved on the first attempt from typed titles. Two structural facts about the harvest, both worth carrying forward:

- **14 of 15 seed pairs share zero references**, max `seed_overlap` 2. Four cluster questions, four separate origins, and no cross-citation between them. The mechanisms were never connected to each other in these fields either.
- **For pre-1990 books the backward channel is dead.** Gibson returned 0 references, Ashby 3, Powers 6, because OpenAlex holds no reference lists for books. The classical half of this review came entirely from forward citations — the exact mirror of §4 of the literature review plan, where the seeds were too *new* for backward hops to reach the benchmark literature.

Cluster D's review literature is Information Systems affordance theory, not ecological psychology.

Three of five D rows are about organisational technology use — *The Materiality of Technology*, *Affordance theory and how to use it in IS research*, *A Systematic Review of Empirical Affordance Studies*. That is not a harvest failure: ecological psychology's overviews are books that OpenAlex indexes without abstracts or reference lists, and the field that writes *indexed reviews* about affordances is IS. One row says so in its own abstract — "the transfer of the affordance concept that originated in ecological psychology to the field of Information Systems gave rise to ontological discussions".

2. One paragraph per cluster

D — Turning knowledge into action · challenges, weakly

The affordance sources agree on one thing that bears on the agent: an affordance is not a property of an object. Chemero states it directly — affordances are "relations between the abilities of animals and features of the environment ... not properties of either the environment or the animal". Davis's *How Artifacts Afford* makes the same move with a condition attached: affordances "mediate between the

features of a technology and the outcomes of engagement", under stated mechanisms and conditions. Read onto the agent, that says a control relation is indexed to the agent's repertoire and to the state it was found in, which is not what a one-off opening phase stores. But every source making this argument is making it about organisational technology, and the one chapter that examines Gibson's own premises does so only to port them into IS research, flagging "three unresolved issues" in the translation. **The challenge is real in form and unsupported in this corpus.**

A — Understanding an unknown world, what is probably true · supports, and reframes

The strongest cluster, and the only one where the sources say something an implementation can use. Gopnik and Meltzoff's theory holds that children "like scientists, learn about the world by forming and revising theories" — the unit that changes is the theory, not the observation. *Bayesian Models of Conceptual Development* formalises that as "inference over hierarchies of probabilistic generative programs". Heckerman's chapter supplies the distinction the agent needs most: a causal reading of a learned network is what "allows us to predict the effects of interventions", and an acausal one does not. Together these support **H1** and **H6** while reframing them — the useful unit is a competing program, not a confidence-tagged assertion, and revision means replacing a rule rather than appending a counter-example.

C — Discovering useful information · nothing

Four rows, four **nothing** verdicts. Conant & Ashby resolved as a seed, and its forward literature in review form is second-order cybernetics applied to systemic therapy, architecture and organisational ethics. Wiener's *Cybernetics* is in the shortlist and its abstract states the programme rather than a mechanism. **The law of requisite variety did not reach this corpus in a form that yields an agent-side quantity.** Getting the floor it promises — a bound on distinct responses needed against distinct environment states — requires the primary text, which the stopping rule excludes. Recorded as *not found here*, not as absent.

E — Improving over experience · challenges, and the most interesting row

One usable source: *Control at the heart of life*, a 2025 philosophical review of perceptual control theory. Its claim is the inversion — behaviour is whatever varies to hold a perceptual variable at a reference value, organised as "the hierarchy of aims or values". The other E row is sociology: identity theory absorbed "the perceptual control system" as its mechanism, with social meaning as the controlled variable, which has no counterpart in the agent's observation space. So PCT's uptake has been in sociology rather than in agent design, and its one review here challenges not **H4** but **the act-then-observe loop that every hypothesis in the register assumes.**

3. What this changes

Marked as the sitting required: `mechanism-candidate` / `argument-against` / `background-only`. Nothing here gets a verdict on its own — candidates become register blocks with falsifiers, like everything else.

`argument-against` — **H3 rests on an assumption this pass could not test.** See §4.

`mechanism-candidate` — **a reference-value harness (from PCT).** The agent declares a perceptual variable and a target value and acts to reduce error, instead of choosing an action and observing the result. "Keep the counter non-zero" survives a reset; "press ACTION2" does not. Baseline: the same frozen model choosing actions directly. Falsified if held-out `max_level_cleared` sits within the baseline's seed-to-seed spread. **This is not H1–H6** — it is a different control architecture, and it is the one thing in this appendix the register has no block for.

`mechanism-candidate` — **score the world model on intervention prediction, not observation fit.** Hold out an action never taken in a state and ask the model what it does. From Heckerman: only the causal reading supports that prediction. A model that fits the action log and fails this test is the specific failure the causal literature names, and the register has no test that would catch it.

`mechanism-candidate` — **competing programs instead of confidence tags.** Hold two incompatible world programs and act to separate them, rather than one model carrying a confidence value. Falsifier: two-program runs reach a level index the single-model baseline never reaches. Reframes H1 rather than replacing it.

`mechanism-candidate` — **repeat-refutation rate as a dense metric.** How many times is a refuted rule re-applied after its refutation? It moves long before a level is solved, it is cheap to compute from the action log, and no current hypothesis names it. Fits the dense-metric promotion already forced by the E0 random-floor pre-check.

`background-only` — Wiener's *Cybernetics*; second-order cybernetics; the development of identity theory; the three IS affordance sources; and four rows that matched on keywords alone (infant-directed speech acoustics, sport participation, nature play, a general-systems reader).

4. The register amendment, as required

Two sentences were pre-specified for the register. Neither is honest as written, because the corpus that would decide between them does not exist here. The amendment is therefore the third form, and belongs in [experiment_register.md](#) under the H3 experiment:

Amendment, 12 Sept 2026 — H3 and the affordance account. The affordance literature holds that a control relation is a relation between an agent's abilities and the environment's features, not a

property of an object (Chemero 2003; Davis 2020). If that transfers, H3's premise — that controllability is identified once, in an opening phase — is **mis-specified rather than wrong**: what an agent can identify is a relation conditional on state and repertoire, and a relation found at level 1 need not hold at level 3. **This is not yet grounds to revise H3**. No source in this corpus engages Gibson's premise directly; cluster D's review literature is Information Systems affordance theory, and the corpus is known to under-sample ecological psychology, whose overviews are books this method cannot reach. Treat as *not found in this corpus*, not as *unclaimed*. **What follows for E2 regardless.** Instrument H3 so the two readings are distinguishable: re-test controllability at every level rather than once per game, and record how often a control relation learned at level n still holds at level $n+1$. If it usually holds, the one-off phase is defensible and the affordance objection does not bite. If it usually does not, a null result for E2 reads as *H3 mis-specified*, not as an implementation failure. That measurement costs one extra field in the run log.

5. Honest limits of this pass

- **Abstract-only.** 1 of 16 PDFs. Every claim above is an abstract-level reading, and two rows carry a source-type judgement with no supporting quote — flagged by the extraction audit and left standing rather than papered over.
- **One row could not be screened at all.** *Theory-based Bayesian models of inductive learning and reasoning* has no abstract indexed; recorded as `not reported` rather than dropped.
- **Cluster C is unresolved, not empty.** The requisite-variety bound is the one quantity in this appendix that could become a metric, and reaching it needs the primary text.
- **The corpus cannot answer the Gibson question.** That is the finding. A pass that wants to answer it should read the two books directly rather than widening this funnel.
- **Outside the active bounded task.** Written during T-E0c, which scopes `experiments`, `src/agents` and `tests`. No code was changed for it.